Effective (co)descent morphisms in some topological categories

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We deal with the problem of whether it is sufficient, when investigating the problem of the effectiveness of a descent morphism, to restrict the consideration only to the descent data (C, γ, ξ), where γ lies in a morphism class $\mathbb{E}$; the notion of a factorization system and the dual to the amalgamation property play key roles in our discussion. With the aid of the obtained results, we find conditions on a functor F preserving (not necessarily all pullbacks, but) pullbacks of morphisms from $\mathbb{E}$, under which the functor F reflects the property "all descent morphisms are effective". The following two particular cases are of special interest: (1) F is a full coreflection; (2) F is a topological functor. Applying the results obtained along this line, we show that all regular monomorphisms are effective codescent morphisms in the category of topological spaces; note that this result was obtained earlier by Mantovani in a different way (unpublished). With the aid of our approach, we also show that all codescent morphisms are effective in the categories of Hausdorff topological spaces, compact Hausdorff topological spaces, normal topological spaces, Banach spaces, metric spaces, complete metric spaces, and some others (in fact, all regular monomorphisms are such in several of these categories). For the internal characterizations of effective codescent morphisms in some of these categories, partial results are obtained. The complete characterization of effective codescent morphisms $p : B \to E$ is given for the category of Hausdorff topological spaces, in the case of compact B. Further, a new proof of the result by Gran and Rossi asserting that all regular epimorphisms are effective descent morphisms in the category of topological $\mathcal{V}$-algebras, where $\mathcal{V}$ is a Mal'cev variety of universal algebras is given.